Tables from EXCEL sheet

	A	B	C	D	E
2	Inter Arrival Time(IAT) Table				
3					
4	IAT	Probability	Cp	lower	Upper
5	1	0.125	0.125	0	125
6	2	0.125	0.25	126	250
7	3	0.125	0.375	251	375
8	4	0.125	0.5	376	500
9	5	0.125	0.625	501	625
10	6	0.125	0.75	626	750
11	7	0.125	0.875	751	875
12	8	0.125	1	876	1000

Table 01: Distribution of Inter Arrival Time Table.

		Service Time		
Service Time	Probability	Cp	Lower	Upper
1	0.1	0.1	0	10
2	0.2	0.3	11	30
3	0.3	0.6	31	60
4	0.25	0.85	61	85
5	0.1	0.95	86	95
6	0.05	1	96	100

Table 02: Distribution of Inter Service Time Table.

Here,

Cumulative Probability (Cp) = Previous CP + Current Probability

= SUM (C5 , B5)

Upper Limit formula = Cp×100

= C5*1000

Upper Limit formula = Previous Upper+1

= E5+1

Table 03 and 04: Compute Arrival time and Service time for Random Digits

Determine Time Between Arrival		
Customer no	Random digit	Time Between Arrival
1	-	-
2	161	2
3	658	6
4	359	3
5	606	5
6	198	2

Determine Service Time		
Customer no	Random Digit	Service Time
1	66	4
2	97	6
3	89	5
4	2	1
5	16	2
6	70	4

Here,

Time Between Arrival = The value whose RN Range contains the Random Digit

=IF(J8<126,1,IF(J8<251,2,IF(J8<376,3,IF(J8<501,4,IF(J8<625,5,IF(J8<751,6,IF(J8<876,7,IF(J8<1001,8))))))

Service Time = The value whose RN Range contains the Random Digit

=IF(O6<11,1,IF(O6<31,2,IF(O6<61,3,IF(O6<86,4,IF(O6<96,5,IF(O6<101,6))))))

Simulation table

Customer	IAT	AT	ST	TSB	TSE	WT	TTS	Idle Time
1	0	0	4	0	4	0	4	0
2	2	2	6	4	10	2	8	0
3	6	8	5	10	15	2	7	0
4	3	11	1	15	16	4	5	0
5	5	16	2	16	18	0	2	0
6	2	18	4	18	22	0	4	0

Table: 05: Simulation Table

Here,

TSB = MAX(M18,J19)

TSE = Sheet1!$K19+Sheet1!$L19

WT = Sheet1!$L19-Sheet1!$J19

TTS = Sheet1!$N19+Sheet1!$K19

Idle Time = =L19-M18

Source Code:

Mount drive

```
import pandas as pd

from google.colab import drive

drive.mount('/content/drive')
```

Create Inter Arrival Time

```
data = { "IAT": list(range(1, 9)),"Probability": [0.125] * 8,}

df = pd.DataFrame(data)

df["Cumulative Probability"] = df["Probability"].cumsum()

df["Upper"] = (df["Cumulative Probability"] * 1000).astype(int)

file_path = "/content/drive/MyDrive/iat_table.csv"

df.to_csv(file_path, index=False)

print(df)

print("CSV file saved to Google Drive:", file_path)
```

IAT	Probability	Cumulative Probability	Upper
1	0.125	0.125	125
2	0.125	0.250	250
3	0.125	0.375	375
4	0.125	0.500	500
5	0.125	0.625	625
6	0.125	0.750	750
7	0.125	0.875	875
8	0.125	1.000	1000

Create Service Time Table

```
data = {"Service Time": [1, 2, 3, 4, 5, 6],"Probability": [0.1, 0.2, 0.3, 0.25, 0.1, 0.05]}

df = pd.DataFrame(data)

df["Cp"] = df["Probability"].cumsum()

df["Upper"] = (df["Cp"] * 100).astype(int)

df = df[["Service Time", "Probability", "Cp", "Upper"]]

file_path = "/content/drive/MyDrive/service_time_table.csv"
```

```
df.to_csv(file_path, index=False)
print(df)
print("CSV file successfully saved at:")
print(file_path)
```

Service Time	Probability	Cp	Upper
1	0.10	0.10	10
2	0.20	0.30	30
3	0.30	0.60	60
4	0.25	0.85	85
5	0.10	0.95	95
6	0.05	1.00	100

<u>Take IAT and service time table as INPUT</u>

```
iat_path = "/content/drive/MyDrive/iat_table.csv"
service_path = "/content/drive/MyDrive/service_time_table.csv"
iat_df = pd.read_csv(iat_path)
service_df = pd.read_csv(service_path)
print("IAT Table:")
display(iat_df)
print("Service Time Table:")
display(service_df)
```

IAT Table:

	IAT	Probability	Cumulative Probability	Upper
0	1	0.125	0.125	125
1	2	0.125	0.250	250
2	3	0.125	0.375	375
3	4	0.125	0.500	500
4	5	0.125	0.625	625
5	6	0.125	0.750	750
6	7	0.125	0.875	875
7	8	0.125	1.000	1000

Service Time Table:

	Service Time	Probability	Cp	Upper
0	1	0.10	0.10	10
1	2	0.20	0.30	30
2	3	0.30	0.60	60
3	4	0.25	0.85	85
4	5	0.10	0.95	95
5	6	0.05	1.00	100

Time Between Arrival

```
data = {"Customer no": [1, 2, 3, 4, 5, 6], "Random digit": [None, 161 ,658, 359,606,198]}

cust_df = pd.DataFrame(data)

def determine_iat(random_digit, iat_table):

    if pd.isna(random_digit):

        return None

for _, row in iat_table.iterrows():

        if random_digit <= row["Upper"]:

            return row["IAT"]

 return None

cust_df["Time Between Arrival"] = cust_df["Random digit"].apply(lambda x: determine_iat(x,
iat_df))

iat_results_df = cust_df[["Customer no","Random digit", "Time Between Arrival"]]

display(iat_results_df)
```

	Customer no	Random digit	Time Between Arrival
0	1	NaN	NaN
1	2	161.0	2.0
2	3	658.0	6.0
3	4	359.0	3.0
4	5	606.0	5.0
5	6	198.0	2.0

service between time

```
data = {"Customer no": [1, 2, 3, 4, 5, 6],"Random digit": [66,97,89,2,16,70]}

cust_df = pd.DataFrame(data)

def determine_service(random_digit, service_time_table):

    for _, row in service_time_table.iterrows():

        if random_digit <= row["Upper"]:

            return row["Service Time"]

    return None

cust_df["Service Between Arrival"] = cust_df["Random digit"].apply(lambda x: determine_service(x,
service_df))

service_results_df = cust_df[["Customer no","Random digit", "Service Between Arrival"]]
```

```
service_results_df = service_results_df.rename(columns={"Random digit": "Random digit (Service)"})

display(service_results_df)
```

Customer no	Random digit (Service)	Service Between Arrival
1	66	4.0
2	97	6.0
3	89	5.0
4	2	1.0
5	16	2.0
6	70	4.0

Simulation Table

```
simulation_table = pd.merge(
    iat_results_df,
    service_results_df[["Customer no", "Random digit (Service)", "Service Between Arrival"]],
    on="Customer no",
    how="left")
print("Merged Simulation Table:")
display(simulation_table)
import pandas as pd
cust_df = simulation_table.copy()
cust_df = cust_df.rename(columns={
    "Customer no": "Customer",
    "Time Between Arrival": "IAT",
    "Service Between Arrival": "ST"
})
cust_df.loc[0, "IAT"] = 0
cust_df["IAT"] = cust_df["IAT"].astype(int)
cust_df["ST"] = cust_df["ST"].astype(int)
cust_df["AT"] = 0
cust_df["TSB"] = 0
cust_df["TSE"] = 0
```

```
cust_df["WT"] = 0
cust_df["TTS"] = 0
cust_df["Idle Time"] = 0
for i in range(len(cust_df)):
if i == 0: cust_df.loc[i, "AT"] = 0
    else: cust_df.loc[i, "AT"] = cust_df.loc[i-1, "AT"] + cust_df.loc[i, "IAT"]
 if i == 0:
 cust_df.loc[i, "TSB"] = 0
    else:
        cust_df.loc[i, "TSB"] = max(cust_df.loc[i, "AT"], cust_df.loc[i-1, "TSE"])
 cust_df.loc[i, "TSE"] = cust_df.loc[i, "TSB"] + cust_df.loc[i, "ST"]
cust_df.loc[i, "WT"] = cust_df.loc[i, "TSB"] - cust_df.loc[i, "AT"]
cust_df.loc[i, "TTS"] = cust_df.loc[i, "WT"] + cust_df.loc[i, "ST"]
if i == 0: cust_df.loc[i, "Idle Time"] = cust_df.loc[i, "TSB"]
    else: cust_df.loc[i, "Idle Time"] = max(0, cust_df.loc[i, "TSB"] - cust_df.loc[i-1, "TSE"])
final_df = cust_df[["Customer","IAT","AT","ST","TSB","TSE","WT","TTS","Idle Time"]]
display(final_df)
```

OUTPUT:

```
final_df
```

	Customer	IAT	AT	ST	TSB	TSE	WT	TTS	Idle Time
0	1	0	0	4	0	4	0	4	0
1	2	2	2	6	4	10	2	8	0
2	3	6	8	5	10	15	2	7	0
3	4	3	11	1	15	16	4	5	0
4	5	5	16	2	16	18	0	2	0
5	6	2	18	4	18	22	0	4	0